Pallavi Pandey

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AI Full Stack Engineer | Backend Engineer | Distributed Systems | Cloud & Generative AI

6+ years of experience designing scalable backend systems, asynchronous FastAPI microservices, distributed APIs, cloud-native applications, Retrieval-Augmented Generation (RAG) systems, and production AI infrastructure across AWS, Azure, and GCP.

Education

B.S. in Data Science and Applications — **Indian Institute of Technology, Madras**

Relevant Coursework

Distributed Systems • Operating Systems • DBMS • Software Engineering • Data Structures & Algorithms • Machine Learning • Deep Learning • Cloud Computing

Professional Experience

AI Full Stack Engineer | Dom Ventas, Bengaluru — ***Jan 2026 - Current***

- Architecting **AI-powered** retail and operations platforms using FastAPI, PostgreSQL, Redis, Docker, and cloud-native services, automating critical workflows across multiple business teams.
- Designing and implementing a **Retrieval-Augmented Generation (RAG)** chatbot using embeddings, vector databases, and LLMs for intelligent document retrieval and conversational assistance.
- Building scalable **ETL pipelines** that consolidate data from multiple databases into unified analytics dashboards for marketing and operations intelligence.
- Developing secure backend APIs with JWT authentication, RBAC, caching, asynchronous processing, and background task execution for high-performance applications.

Lead Software Engineer | Articence Inc, CA — ***Jan 2025 - Jan 2026***

- Migrated backend architecture from **Flask to FastAPI**, improving performance, enabling asynchronous processing, and enhancing modularity and maintainability.
- Integrated advanced AI capabilities, including **LLMs, RAG, Text-to-Speech (TTS), and Speech-to-Text (STT)**, to expand intelligent automation features.
- Implemented **WebSockets and webhooks** to enable real-time, bidirectional communication between client and server.

AI Engineer | Magizh Technologies, Coimbatore — ***Jan 2024 - Dec 2024***

- Supported backend development of a scalable educational platform for competitive exam preparation, enabling structured content delivery and resource management.
- Developed and integrated an **AI-driven chatbot** providing real-time assistance and personalized learning recommendations to boost engagement.
- Implemented **microservices** for OAuth and multi-channel notifications (Slack, SMS, LinkedIn) to enhance authentication and communication.

Python Developer | IIT Madras, Chennai — ***Jan 2023 - Sept 2023***

- Developed the **"Best Project"** feature on the IITM BS Portal, enabling instructors to add and manage top student projects for each term.
- Built a **Paradox'23 Showcase** page displaying highlights of IIT Madras annual fest to enhance engagement.
- Designed responsive, user-friendly pages to ensure seamless access and improved portal experience.

Junior Developer | VaayuSastra, IITM Research Park, Chennai — ***Aug 2022 - Oct 2023***

- Collaborated in a team to develop **Vayu Shastra**, a learning app for students enrolled in their courses.
- Built an Android application using **Flutter** for a smooth, cross-platform user experience.
- Integrated **MongoDB** for backend database management and data storage.

Technical Skills

- **Programming Languages:** Python, SQL, JavaScript, TypeScript, GraphQL, HTML5, CSS, Java (Basics), Bash/ Shell Scripting
- **Backend & Frameworks:** FastAPI, Flask, Django, Django REST Framework (DRF), Express.js, Node.js, Next.js, Vue.js, React (Basics)

- **AI & Machine Learning:** LLMs, RAG, AI Agents, LangChain, Hugging Face, NLP, Prompt Engineering, Embeddings, Fine-tuning, Vector DB, Pinecone, FAISS, ChromaDB, OpenAI APIs, MCP, Tool Calling, SciPy
- **Architecture & APIs:** Microservices, Event-Driven Architecture, RESTful APIs, GraphQL APIs, MVC, RBAC, Serverless, Multi-tenant Architecture, Clean Architecture
- **Databases & Caching:** PostgreSQL, MongoDB, MySQL, Oracle, Elasticsearch, Redis, DynamoDB, Firebase Firestore, Neo4j (Graph DB), TimescaleDB
- **Cloud & Infrastructure:** AWS (EC2, Lambda, RDS, S3), Azure, GCP, Cloud Functions, Cloud Run, Pub/Sub, Kubernetes, Terraform, PySpark, Distributed Systems
- **Libraries & Data Tools:** Pandas, NumPy, Streamlit, Matplotlib, Seaborn, Requests, BeautifulSoup, Selenium (Web Scraping), Asyncio, Pydantic
- **DevOps & CI/CD:** Docker, Git, GitHub, GitLab, GitHub Actions, GitLab CI, Jenkins, Helm, Postman, CI/CD Pipeline Building, Prometheus, Grafana, ELK Stack
- **Messaging, Security & Backend Tools:** Kafka, RabbitMQ, Celery, WebSockets, Webhooks, Nginx, Gunicorn, OAuth2, JWT, Authentication & Authorization, Rate Limiting, Caching, Load Balancing, Pytest, Unit Testing

Training & Workshops

Distributed Computing Faculty | JAIN (Deemed-to-be University), Jayanagar, Bengaluru

- Teaching Module 1 hands-on labs to **M.Tech CSE** students — sockets, RPC/gRPC, concurrency, mutual exclusion, and deadlocks.
- Delivering Module 2 on **fault tolerance, consistency/CAP theorem**, and the security stack from authentication through auditing.

Workshop: AI × Cybersecurity | Jain University, Global Campus, Kanakapura

- Delivered 4 hours of theory on **AI-powered threat detection, prompt injection, and AI-system security**.
- Led 4 hours of hands-on labs on **anomaly-based threat detection, incident response, and SOC alert triage**.

Placement Training | CMS Business School, Gandhi Nagar, Bengaluru

- Trained **2nd Year MBA students** on Aptitude — arithmetic shortcuts, percentages, ratios, and calculation speed techniques.
- Trained students on **Advanced Excel** — VLOOKUP/XLOOKUP, INDEX-MATCH, SUMIFS/COUNTIFS, and PivotTables/PivotCharts.

Artificial Intelligence Faculty (Semester 5) | St. Claret College Autonomous, Bengaluru

- Taught **Deep Learning and Computer Vision** — neural networks, backpropagation, CNNs, and object detection (RCNN/YOLO).
- Taught **Forecasting and Advanced NLP** — RNN/LSTM/GRU and Transformer architectures, with a capstone mini project.

Data Analysis Training Program | Dr. Ambedkar Institute of Technology, Bengaluru

- Trained engineering students on **Data Analysis** using SQL, Python, Pandas, and NumPy.
- Covered **Power BI** dashboarding to build industry-ready data analysis skills.

Guest Lecture: AI & Data Science | VIT, Vellore

- Delivered a guest lecture on the evolving landscape of **AI and Data Science** to engineering students.
- Bridged academic theory with real-world industry practice and applications.

Workshop: Blockchain & Web3 Fundamentals | Magizh Technologies, Coimbatore

- Trained developers on **Blockchain fundamentals** and smart contract development with Solidity.
- Covered **DeFi concepts** and hands-on deployment on Ethereum testnets using Hardhat and Web3.js.

Training: Generative AI in Practice | Bengaluru, Whitefield

- Trained **B.Tech interns** on LLM fundamentals and RAG pipelines.
- Covered **prompt engineering** and agentic workflows with LangChain.

Co-Curricular, Extra-Curricular Activities & Hobbies

- Participated in the **Kharagpur Data Science Hackathon 2024** organized by **IIT Kharagpur**, focusing on data-driven problem-solving and model development.
- Competed in **HackCBS 6.0** (2023), **India's largest student-run hackathon**, collaborating in a team to build innovative technology solutions under time constraints.
- Passionate about dance and performing arts; performed in dance and drama at IIT **Madras's** annual fest, **Paradox**.